

1. User Service — user_db

users

Field	Type	Key	Description
id	UUID	PK	User ID
first_name	VARCHAR(100)		First name
last_name	VARCHAR(100)		Last name
email	VARCHAR(255) UNIQUE		Login email
password	VARCHAR(255)		Encrypted password
phone	VARCHAR(20)		Phone number
role	VARCHAR(30)		CUSTOMER / ADMIN
status	VARCHAR(30)		ACTIVE / BLOCKED
created_at	TIMESTAMP		Created time
updated_at	TIMESTAMP		Updated time

addresses

Field	Type	Key
id	UUID	PK
user_id	UUID	FK → users.id
address_line1	VARCHAR(255)	
address_line2	VARCHAR(255)	

Field	Type	Key
city	VARCHAR(100)	
state	VARCHAR(100)	
country	VARCHAR(100)	
pincode	VARCHAR(20)	
address_type	VARCHAR(30)	
is_default	BOOLEAN	
created_at	TIMESTAMP	

Relationship

users

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addresses

One User → Many Addresses

2. Product Service — product_db

categories

Field	Type	Key
id	UUID	PK
name	VARCHAR(100)	UNIQUE
description	TEXT	

Field	Type	Key
status	VARCHAR(30)	
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

products

Field	Type	Key
id	UUID	PK
category_id	UUID	FK → categories.id
name	VARCHAR(255)	
sku	VARCHAR(100)	UNIQUE
description	TEXT	
brand	VARCHAR(100)	
price	NUMERIC(15,2)	
currency	VARCHAR(3)	
status	VARCHAR(30)	
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

product_images

Field	Type	Key
id	UUID	PK
product_id	UUID	FK → products.id
image_url	VARCHAR(500)	
is_primary	BOOLEAN	
created_at	TIMESTAMP	

Relationships

categories

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products

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product_images

One Category → Many Products

One Product → Many Images

3. Inventory Service — inventory_db

inventory

Field	Type	Key
id	UUID	PK
product_id	UUID	UNIQUE
available_quantity	INTEGER	
reserved_quantity	INTEGER	
sold_quantity	INTEGER	
reorder_level	INTEGER	
version	BIGINT	
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

product_id comes from **Product Service**.

There is **no cross-database FK**.

inventory_reservations

Field	Type	Key
id	UUID	PK
order_id	UUID	
product_id	UUID	
quantity	INTEGER	
status	VARCHAR(30)	
reserved_at	TIMESTAMP	
released_at	TIMESTAMP	
created_at	TIMESTAMP	

Relationship inside Inventory DB:

inventory

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inventory_reservations

Conceptually:

Product Service

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| product_id



Inventory Service

4. Order Service — order_db

This is the most important database.

orders

Field	Type	Key
id	UUID	PK
order_number	VARCHAR(50)	UNIQUE
user_id	UUID	
status	VARCHAR(30)	

Field	Type	Key
total_amount	NUMERIC(15,2)	
currency	VARCHAR(3)	
shipping_address	JSONB	
billing_address	JSONB	
created_at	TIMESTAMP	
updated_at	TIMESTAMP	

user_id comes from User Service.

order_items

Field	Type	Key
id	UUID	PK
order_id	UUID	FK → orders.id
product_id	UUID	
product_name	VARCHAR(255)	
sku	VARCHAR(100)	
quantity	INTEGER	
unit_price	NUMERIC(15,2)	
total_price	NUMERIC(15,2)	
created_at	TIMESTAMP	

Relationship

orders

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order_items

One Order → Many Order Items

5. Payment Service — payment_db

payments

Field	Type	Key
id	UUID	PK
order_id	UUID	UNIQUE
user_id	UUID	
payment_reference	VARCHAR(100)	UNIQUE
amount	NUMERIC(15,2)	
currency	VARCHAR(3)	
payment_method	VARCHAR(30)	
status	VARCHAR(30)	
updated_at	TIMESTAMP	
created_at		

payment_transactions

Field	Type	Key
id	UUID	PK

Field	Type	Key
payment_id	UUID	FK → payments.id
transaction_reference	VARCHAR(100)	UNIQUE
gateway	VARCHAR(50)	
gateway_transaction_id	VARCHAR(100)	
amount	NUMERIC(15,2)	
status	VARCHAR(30)	
failure_reason	TEXT	
created_at	TIMESTAMP	

Relationship

payments

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payment_transactions

One Payment → Many Transactions

Useful when a payment has retries or gateway attempts.

6. Notification Service — notification_db

notifications

Field	Type	Key
id	UUID	PK
user_id	UUID	
order_id	UUID	
type	VARCHAR(50)	
channel	VARCHAR(20)	
recipient	VARCHAR(255)	
subject	VARCHAR(255)	
message	TEXT	
status	VARCHAR(30)	
retry_count	INTEGER	
created_at	TIMESTAMP	
sent_at	TIMESTAMP	

No FK to User or Order because those belong to other services.